SPACECRAFT & GROUND STATION TT&C IP CORE DOCUMENTATION

CCSDS-SAT-TTC-IP (v2.39.0): Product Guide & User Manual

Synthesizable IEEE 1800-2017 SystemVerilog Transceiver IP Core
for Spacecraft Flight (`ROLE_MODE=0`), Ground EGSE (`ROLE_MODE=1`),
and Universal Crosslink / On-Orbit BIST (`ROLE_MODE=2`)

Document Attribute	Specification Value
Document ID	`PG-CCSDS-TTC-v2.39.0`
Top-Level RTL Module	`ccsds_sat_ttc_top` (`rtl/ccsds_sat_ttc_top.sv`)
Core Version ID Register	`32'hCC5D_0230` (`CORE_ID_VER` at offset `0x00`)
Supported CCSDS Standards	17 Blue/Orange Books (TM, TC, AOS, Proximity-1, USLP, COP-1/P, SDLS, PN, RF)
Verification Sign-Off	Stages 0–12 Verified with **0 Errors** in Xilinx Vivado 2022.2 `xsim`
Measured True $F_{\max}$ (Vivado STA)	`clk_sys`: **116.40 MHz** | `clk_adc_iq`: **135.41 MHz** | `clk_tx_sym`: **169.46 MHz**
Flight Software SDK	Bare-Metal ISO C11 HAL (`sw/ccsds_ttc_hal.h`, `sw/ccsds_ttc_hal.c`)
License	Apache License 2.0

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1 Product Overview & Architectural Personas

The **CCSDS-SAT-TTC-IP (ccsds_sat_ttc_top)** is a radiation-mitigated, vendor-agnostic **Telemetry, Tracking, and Command (TT&C)** digital baseband, channel coding, data link framing, closed-loop ARQ, and inline cryptographic security IP core written in strict synthesizable IEEE 1800-2017 SystemVerilog. It implements **17 CCSDS Blue and Orange Book standards** within a single parameterized RTL hierarchy:

- **Spacecraft Flight Transceiver Persona (ROLE_MODE = 0, 23 Synthesizable Modules):** Implements the complete spacecraft uplink command receiver and downlink telemetry transmitter while stripping ground-only decoders and retransmission queues to minimize radiation-hardened FPGA area and Single-Event Upset (SEU) cross-section.
- **Synthesizable Hardware Ground EGSE Persona (ROLE_MODE = 1, 22 Synthesizable Modules):** Instantiates the 17 shared core modules plus the 5 ground-segment endpoints (`ccsds_fop1_engine`, `ccsds_tc_cltu_framer`, `ccsds_bch6356_enc`, `ccsds_dl_rs255_223_dec` full $t = 16$ Berlekamp-Massey/Chien/Forney error-correcting decoder, and `ccsds_tm_aos_deframer`) for FlatSat benches, RFSoC ground station modems, and Hardware-in-the-Loop (HIL) test equipment.
- **Universal Crosslink & On-Orbit BIST Persona (ROLE_MODE = 2, All 28 Synthesizable Modules):** Instantiates both spacecraft flight and ground EGSE endpoints simultaneously for symmetric spacecraft-to-spacecraft Proximity-1 (`CCSDS 211.0-B-6`) and USLP (`CCSDS 732.1-B-3`) crosslinks as well as autonomous on-orbit Built-In Self-Test (BIST).

1.1 Summary of Supported CCSDS Standards

CCSDS Standard	Protocol / Layer	Implemented Hardware Capabilities in `ccsds_sat_ttc_top`
133.0-B-2	Space Packet Protocol	48-bit Primary Header validation, Cycle-0 FHP segmentation, APID=0x7FF Idle Packet generation & automatic Rx stripping.
132.0-B-3	TM Space Data Link	6-octet TM Header, MC/VC frame counters, 8-byte CUC OBT Secondary Header, atomic 32-bit CLCW OCF, CRC-16 FECF.
232.0-B-4	TC Space Data Link	5-octet TC Header, 8-bit Segment Header (MAP_ID), multi-frame CLTU 0x55 fill stripping, Type-AD/BD/BC frame handling.
232.1-B-2	COP-1 (FARM-1 & FOP-1)	3x TMR FARM-1 responder (8 VCs, Two-Phase SDLS commit, Type-BC Unlock/Set V(R)) & Ground FOP-1 Go-Back-N initiator.
732.0-B-4	AOS Space Data Link	6-octet AOS Header with shortened RS$(10, 6)$ over $GF(2^4)$ FHEC (ccsds_gf256_math.sv) SEC-DED header protection.
732.1-B-3	Unified Space Link (USLP)	Truncated (4B) and Extended (7–14B) Primary Headers, 16-bit SCID, TFDFH construction rules, CRC-16 or CRC-32 FECF.
211.0/1/2-B	Proximity-1 Crosslink	5-octet Version-3 Header, 24-bit ASM (0xFAF320), 32-bit PLCW COP-P ARQ controller, and 32-bit CRC-32 (0x00A00805).
131.0-B-5	TM Sync & Channel Coding	Dual-Basis RS$(255, 223)$ ($I \in \{1..8\}$, implicit $Q \in [0..200]$, $t = 16$ BM/Forney decoder), Rate-$1/2$, $K = 7$ Conv + 64-State Soft Viterbi, 16-State RSC Turbo ($r = 1/2..1/6$), h_{TM} (0xA9) scrambler.
231.0-B-4	TC Sync & Channel Coding	CLTU Start (0xEB90), Expurgated BCH$(63, 56)$ SEC-DED ($g(x) = x^7 + x^6 + x^2 + 1$), Tail (0xC5..79), h_{TC} (0x5F), 3x TMR HCPD.
231.1-O-1	Uplink Short-Block LDPC	Table 2-1/2-4 Quasi-Cyclic $(128, 64)$ LDPC encoder & folded 16-row/clk Soft-Decision Offset Min-Sum (OMS) iterative decoder.
301.0-B-4	CUC On-Board Time (OBT)	48-bit Unsegmented Time Code (32-bit Coarse Seconds + 16-bit Fine Sub-Seconds, P-Field 0x2E) synchronized to 1-PPS.
355.0/1-B	Space Data Link Security	Synthesizable NIST FIPS-197 AES-128-CTR & NIST SP 800-38D AES-128-GCM ($GF(2^{128})$ GHASH 128-bit Tag) + 64-SA SECDED RAM.
401.0-B-32	RF & Modulation Systems	BPSK, QPSK, OQPSK, PCM/PM; NRZ-L/M, Bi-ϕ-L; 4-SPS 17-Tap SRRC ($\alpha = 0.35$); 4-SPS Gardner STR; FLL-Assisted Costas; S $(240/221)$, X $(880/749)$, Ka-B $(3344/749)$ & Ka-A $(3360/749)$ turnaround.
414.1-B-3	Pseudo-Noise (PN) Ranging	6-component weighted-voting T2B and T4B regenerative ranging correlator ($L = 1,009,470$ chips) & clean Q-channel overlay.

2 Compile-Time Parameters, Resource Utilization & Timing Closure

2.1 Top-Level Synthesis Generics (`rtl/ccsds_sat_ttc_top.sv`)

2.2 Measured Xilinx Vivado 2022.2 Resource Utilization & Device Sizing

Out-of-Context (OOC) synthesis in Xilinx Vivado 2022.2 confirms **zero Block RAMs (`0 RAMB36`)** and **zero DSP slices (`0 DSP48E1/E2`)** across all configurations:

2.3 Two-Pass Static Timing Analysis (STA) & Measured True $F_{\max}$

Parameter Name	Default	Values	Hardware Synthesis Behavior
ROLE_MODE	0	0, 1, 2	0: Spacecraft Flight Core; 1: Hardware Ground EGSE; 2: Universal Crosslink / On-Orbit BIST Transceiver.
ENABLE_SDLS	1'b1	1'b0, 1'b1	When 1'b1, instantiates ccsds_sdls_engine (AES-128-CTR/GCM + 64-SA SECDED RAM). Setting 1'b0 bypasses SDLS and saves **7,919 LUTs** and **5,571 FFs**.
ENABLE_UL_LDPC	1'b1	1'b0, 1'b1	When 1'b1, instantiates ccsds_ul_short_ldpc_dec (231.1-O-1 Soft-OMS LDPC decoder). Setting 1'b0 for BCH(63, 56)-only missions saves **12,610 LUTs**.
ENABLE_DL_TURBO	1'b1	1'b0, 1'b1	When 1'b1, instantiates ccsds_dl_turbo_ldpc_enc. Setting 1'b0 for RS(255, 223)/Convolutional-only downlinks elides the Turbo/LDPC encoder.

Configuration Profile	Slice LUTs	Slice FFs	Zynq-7010 (xc7z010)	Zynq-7020 (xc7z020)	Kintex UltraScale (xcku040 / xqrku060)
Full Flight (SDLS=1, LDPC=1)	**28,861**	**13,864**	163.98%	**54.25%**	**11.91% / 8.70%**
BCH + GCM (SDLS=1, LDPC=0)	**∼16,251**	**∼12,732**	**92.34%**	**30.55%**	**6.70% / 4.90%**
Compact (SDLS=0, LDPC=0)	**∼8,332**	**∼7,161**	**47.34%**	**15.66%**	**3.44% / 2.51%**

Clock Domain	Nominal Target	Pass 1 WNS	Min Period ($T_{\min}$)	Measured True $F_{\max}$	Levels	Limiting Physical Critical Path (Vivado 2022.2)
clk_sys	100.0 MHz (10.000 ns)	**+1.409 ns**	8.591 ns	**116.40 MHz** (+16.40%)	**7**	u_rx_commit_cdc_fifo/rd -> u_csr/cnt_uncorr_err_q_
clk_adc_iq	120.0 MHz (8.333 ns)	**+0.948 ns**	7.385 ns	**135.41 MHz** (+12.84%)	**9**	u_phy_modem/rx_phase_q1 -> u_phy_modem/i_derot_q_r
clk_tx_sym	50.0 MHz (20.000 ns)	**+14.099 ns**	5.901 ns	**169.46 MHz** (+238.9%)	**9**	u_tx_hwm_cdc_fifo/wr_bi -> u_tx_hwm_cdc_fifo/rd_gr
clk_tc_safe	16.0 MHz (62.500 ns)	**+53.427 ns**	9.073 ns	**110.22 MHz** (+588.9%)	**11**	u_ul_deframer/cur_tc_hd -> u_farm1/clcw_scheduled_

3 Top-Level Signal Interface & AXI4-Stream Timing

3.1 Complete Pinout Reference (ccsds_sat_ttc_top)

Port Name	Dir	Width	Clock	Functional Description
clk_sys	In	1	clk_sys	System packet, framing, CSR & outer FEC clock (100 MHz nominal).
clk_tc_safe	In	1	clk_tc_safe	Radiation-safe telecommand & emergency HCPD clock (16 MHz nominal).
clk_tx_sym	In	1	clk_tx_sym	Downlink symbol & inner coding clock (50 MHz nominal).
clk_adc_iq	In	1	clk_adc_iq	Baseband RF ADC/DAC & PN ranging clock (120 MHz nominal).
rst_n	In	1	Async	Active-low asynchronous reset (hold low ≥ 8 clk_tc_safe cycles).

Port Name	Dir	Width	Clock	Functional Description
s_axi_awvalid / awready	In/Out	1	clk_sys	AXI4-Lite write address valid/ready handshake.
s_axi_awaddr	In	8	clk_sys	AXI4-Lite write byte address (0x00..0x24).
s_axi_wvalid / wready	In/Out	1	clk_sys	AXI4-Lite write data valid/ready handshake.
s_axi_wdata / wstrb	In	32, 4	clk_sys	32-bit write data and 4-bit byte lane strobe (4'hF).
s_axi_bvalid / bready / bresp	Out/In	1, 1, 2	clk_sys	AXI4-Lite write response (2'b00 = OKAY).
s_axi_arvalid / arready / araddr	In/Out	1, 1, 8	clk_sys	AXI4-Lite read address channel (0x00..0x24).
s_axi_rvalid / rready / rdata / rresp	Out/In	1, 1, 32, 2	clk_sys	AXI4-Lite 32-bit read data channel (2'b00 = OKAY).
link_proto_override_en	In	2	clk_sys	2'b00: Use CSR PROTO_MODE; non-zero overrides protocol mode.
ul_fec_override_en	In	2	clk_tc_safe	2'd0: Direct Frame; 2'd1: BCH$(63,56)$ CLTU; 2'd2: LDPC$(128,64)$.
sdls_en_cfg / bc_require_sdls_cfg	In	1, 1	clk_sys	Mission security straps OR-ed with CSR bits [26] and [27].
pn_ranging_en_cfg / pn_t4b_mode_cfg	In	1, 1	clk_adc_iq	Enable T4B (1) or T2B (0) regenerative PN ranging.
turnaround_xband_cfg	In	1	clk_adc_iq	Selects S-Band (240/221, 0) or X-Band (880/749, 1) when Ka-Band is 0.
mod_mode_cfg / line_code_cfg	In	2, 2	clk_adc_iq	Modulation (0..3) & Line Coding (0..2) mode straps.
rx_i_adc_in / rx_q_adc_in	In	12s, 12s	clk_adc_iq	Signed 12-bit baseband I/Q ADC samples.
rx_iq_valid_in	In	1	clk_adc_iq	Sample valid strobe (supports divided symbol rates div=1,2,4,8).
s_axis_ul_sym_tvalid/tready/tdata/tlast	In/Out	1, 1, 8, 1	clk_tc_safe	Uplink soft-LLR symbol (tdata[3:0]) or direct frame byte input.
m_axis_rx_pkt_tvalid/tready/tdata/tlast	Out/In	1, 1, 8, 1	clk_sys	Reassembled telecommand Space Packet stream to OBC.
m_axis_rx_pkt_tuser	Out	64	clk_sys	Packed ccsds_tuser_t metadata (vcid, map_id, crc_err, sdls_mac_err).
s_axis_tx_pkt_tvalid/tready/tdata/tlast	In/Out	1, 1, 8, 1	clk_sys	Telemetry Space Packet ingress from OBC (tlast on final packet byte).
s_axis_tx_pkt_tuser	In	64	clk_sys	Ingress metadata (tuser.vcid[2:0] selects target Virtual Channel 0..7).
m_axis_dl_cad_tvalid/tready/tdata/tlast	Out/In	1, 1, 8, 1	clk_tx_sym	Scrambled & outer-FEC encoded Downlink CADU byte stream.
tx_i_dac_out / tx_q_dac_out / tx_iq_valid_out	Out	12s, 12s, 1	clk_adc_iq	Signed 12-bit baseband I/Q DAC outputs and sample valid strobe.
hcpd_obc_reset_pulse / safe_mode_force	Out	1, 1	clk_tc_safe	Direct hardware OBC reset pulse & safe-mode force latch (VCID=63).
hcpd_rf_chain_select	Out	2	clk_tc_safe	Redundant RF chain relay select outputs from TMR HCPD.
pn_All_Locked_out	Out	1	clk_adc_iq	High when all 6 PN ranging sequences ($C_1..C_6$) are locked.
fault_irq_out / edac_scrub_req_out	Out	1, 1	clk_sys	Uncorrectable fault IRQ & FPGA configuration SEU scrub request.

4 AXI4-Lite Control & Status Register Map (ccsds_axi_lite_csr.sv)

All writable mission configuration registers (0x04, 0x08, 0x0C) are implemented as **3x Triple Modular Redundant (TMR)** register banks with continuous single-cycle majority voting and autonomous SEU scrubbing.

Offset	Register Name	Access	Reset Value	Bitfields & Hardware Description
0x00	CORE_ID_VER	RO	0xCC5D_0230	[31:16] Magic ID (0xCC5D), [15:0] Version (0x0230).
0x04	CTRL_SCID_PROTO	R/W (TMR)	0x0131_5155	[9:0] SCID_10B (0x155); [14:12] PROTO_MODE (0=TM, 1=AOS, 2=Prox-1, 3=USLP, 4=TC); [19:16] FEC_MODE (0=RS, 1..4=Turbo); [20] OCF_EN; [21] FECF_EN; [22] FECF_CRC32; [23] AOS_FHEC_EN; [24] TC_SEG_HDR_EN; [25] OBT_SEC_HDR_EN; [26] SDLS_EN; [27] BC_REQ_SDLS; [28] SRRC_4SPS_EN; [29] GARDNER_TED_EN; [31:30] TURNAROUND_KABAND (0=S/X, 1=Ka Cat-B 3344/749, 2=Ka Cat-A 3360/749).
0x08	CTRL_USLP_RS_CFG	R/W (TMR)	0x0001_4255	[15:0] USLP_SCID_16B (0x4255); [19:16] RS_INTERLEAVE_I ($I \in \{1..8\}$); [27:20] RS_VIRT_FILL_Q ($Q \in [0..200]$); [28] SDLS_GCM_MODE (1=128-bit AES-128-GCM GHASH); [29] FLL_ASSIST_EN (1=Wide-CFO FLL-Assisted Costas).
0x0C	CTRL_COP1_TFDF_CFG	R/W (TMR)	0x010D_F010	[7:0] WINDOW_HALF_W ($W = 16$); [18:8] TFDF_BYTE_LEN (223); [24] FAULT_IRQ_EN.
0x10	OBT_COARSE_SEC	R/W	0x6530_0000	[31:0] CCSDS 301.0-B-4 CUC On-Board Time Coarse Seconds.
0x14	OBT_FINE_SUBSEC	R/W	0x0000_0000	[15:0] CCSDS 301.0-B-4 CUC Fine Sub-Seconds (2^{-16} s).
0x18	EDAC_CORR_CNTS	RO / W1C	0x0000_0000	[15:0] cnt_secded_corr_q (W1C bit 0); [31:16] cnt_tmr_seu_corr_q (W1C bit 16).
0x1C	FAULT_ERR_CNTS	RO / W1C	0x0000_0000	[15:0] cnt_uncorr_err_q (W1C bit 0); [31:16] cnt_sdls_mac_err_q (W1C bit 16).
0x20	LATEST_CLCW_MON	RO	0x0000_0000	[31:0] Latest 32-bit COP-1 CLCW ([7:0] = $V(R)$ report_value).
0x24	LATEST_PLCW_MON	RO	0x0000_0000	[31:0] Latest 32-bit Proximity-1 COP-P PLCW.

5 Bare-Metal C11 Flight/EGSE HAL SDK Integration

The C11 Hardware Abstraction Layer (sw/ccsds_ttc_hal.h, sw/ccsds_ttc_hal.c) provides type-safe APIs for initializing the transceiver, configuring all protocol/FEC/PHY/SDLS modes, synchronizing CUC On-Board Time, and unpacking live 32-bit CLCW/PLCW status words:

```
#include "ccsds_ttc_hal.h"

int main(void) {
   ccsds_ttc_dev_t dev;
   ccsds_ttc_init(&dev, 0x43C00000u, my_axi_read32, my_axi_write32);

   ccsds_ttc_config_t cfg = {
      .scid_10b = 0x155u,
      .uslp_scid_16b = 0x4255u,
      .proto_mode = CCSDS_PROTO_TM,
      .fec_mode = CCSDS_FEC_RS255_223,
      .ocf_en = true,
      .fecf_en = true,
      .tc_seg_hdr_en = true,
      .obt_sec_hdr_en = true,
      .sdls_en = true,
```

```
        .sdls_gcm_mode = true,
        .srrc_4sps_en = true,
        .gardner_ted_en = true,
        .fll_assist_en = true,
        .turnaround_kaband = CCSDS_TURNAROUND_KA_CAT_B_3344_749,
        .rs_interleave_i = 1u,
        .rs_virt_fill_q = 195u,
        .cop1_window_w = 16u,
        .tfdf_byte_len = 16u,
        .fault_irq_en = true
    };
    ccsds_ttc_configure(&dev, &cfg);
    return 0;
}
```

5.1 Firmware Configuration Gating & 12-Thread 5,760-Config RF Matrix (v2.39.0)

Before writing hardware CSRs (0x04..0x0C), ccsds_ttc_hal_configure() invokes ccsds_ttc_hal_validate_co to enforce **8 CCSDS Blue Book configuration gating rules**:

- **Rule 1 (CCSDS_HAL_ERR_MOD_LINE_INCOMPAT = -10):** Rejects QPSK/OQPSK paired with Manchester Bi-ϕ-L (CCSDS 401.0-B-32; only 10 of 12 waveform pairs are valid).
- **Rule 2 (CCSDS_HAL_ERR_SRRC_BIPHI_INCOMPAT = -11):** Rejects 4-SPS SRRC filtering (srrc_4sps_en=1) with Bi-ϕ-L.
- **Rule 3 (CCSDS_HAL_ERR_GARDNER_REQUIRES_4SPS = -12):** Rejects gardner_ted_en=1 when srrc_4sps_en=0.
- **Rule 4 (CCSDS_HAL_ERR_PROTO_FEC_INCOMPAT = -13):** Rejects Uplink PROTO_TC paired with Downlink-only RS(255,223)/Turbo, Downlink PROTO_TM/AOS paired with Uplink-only BCH(63,56)/LDPC_UL, and PROTO_PROX1 paired with RS/Turbo.
- **Rule 5 (CCSDS_HAL_ERR_CRC32_PROTO_INCOMPAT = -14):** Rejects 32-bit FECF (fecf_crc32=1) on TM/AOS/TC and requires it on PROX1.
- **Rule 6 (CCSDS_HAL_ERR_AOS_FHEC_INCOMPAT = -15):** Rejects aos_fhec_en=1 unless protocol is AOS or USLP.
- **Rules 7 & 8 (-16, -17):** Enforces $I \in \{1,2,3,4,5,8\}$, $Q \leq 215$, TFDF_LEN $\in [1..1024]$, and $W \in [1..128]$.

All **5,760 valid end-to-end system configurations** (10 Waveforms$\times$4 Turnaround Bands$\times$3 Uplink Packet Sizes $\{8, 14, 28$ B$\}\times$ 4 Downlink Frame/OBT/Rand Modes $\{10, 16, 16{+}$OBT$, 32$ B$\}\times$2 Uplink FECs $\{$BCH$(63,56)$, Short-LDPC$(128,64)\}\times$ 3 Downlink Protocols $\{$TM, AOS, USLP$\}$ $\times$ 2 Security Modes $\{$Clear, SDLS$\}$) are verified across 12 parallel xsim threads via .\scripts\run_stage11_matrix_parallel.bat 12 [MC_ITERS] [IMPAIRMENT_MODE].

Per **ECSS-E-ST-50-05C** and **CCSDS 401.0-B**, Stage 11 wires ccsds_space_flight_channel_model into both Uplink and Downlink 16-bit AD9361 I/Q buses (with 4 continuously advancing CLT Gaussian AWGN LFSRs that never reset across trials) and steps each configuration through **6 progressive channel profiles**:

- **P0_CLEAN_REF:** Baseline thermal AWGN (sigma=4) reference.
- **P1_OAT_AWGN_FEC:** Isolated low-SNR Gaussian AWGN (sigma=24..32) + deterministic Baseband I/Q symbol error injection (mask=0xA5), verifying active BCH(63,56) SEC, Short-LDPC(128,64) Soft-OMS, and RS(255,223) ($t = 16$) error correction.
- **P2_OAT_DOPPLER_PN:** Isolated dynamic S-curve Doppler (+-20..24 step), Doppler acceleration (rate=+-2), and Wiener oscillator phase noise.
- **P3_OAT_DELAY_MP:** Isolated slant-range propagation delay (1..7 samples) and Rician specular multipath echo (-18..-30 dB).
- **P4_OAT_RF_ANALOG:** Isolated SSPA soft-compression, I/Q gain/quadrature skew, and Zero-IF ADC DC offset (+-5..6).
- **P5_COMBINED_MC:** All 6 impairments active simultaneously with randomized per-trial signs, delays, payloads, and continuous AWGN seeds.

Each trial is judged against **7 hardware pass/fail criteria**: (1) 16-bit RF path enforcement (`ul_sym_tvalid==0`), (2) Costas/timing lock & ASM correlation, (3) FEC error correction without abort, (4) hardware `CRC-16-CCITT` pass, (5) `SDLS` MAC & anti-replay verification, (6) bit-exact Space Packet payload reassembly (`BER=0, FER=0`), and (7) closed-loop `CLCW V(R)=1` turnaround at Ground EGSE CSR `0x20`.

Measured 12-Thread 34,560-Trial (5,760 Configs × 6 Progressive Profiles) Full Monte Carlo Sign-Off (v2.39.0): Executing `.\scripts\run_stage11_matrix_parallel.bat 12 6 progressive` in Xilinx Vivado 2022.2 `xsim` verified **all 5,760 / 5,760 valid CCSDS RF configurations across all 6 progressive channel impairment profiles (P0_CLEAN_REF, P1_OAT_AWGN_FEC, P2_OAT_DOPPLER_PN, P3_OAT_DELAY_MP, P4_OAT_RF_ANALOG, P5_COMBINED_MC = 34,560 / 34,560 bidirectional trials, 100.0% pass, 0 failures)** in **1,984.46 s (33.07 min) wall-clock time**, transferring **69,120 RF packets** (`576,000 B` Uplink + `639,360 B` Downlink Space Packet payload = `1,215,360 B` across **39,306,240 16-bit AD9361 RF I/Q samples**: `19,077,120` UL + `20,229,120` DL) and correcting **all 23,040 pre-FEC injected channel bit errors to 0 post-FEC bit errors (BER** $= 0.0$**, FER** $= 0.0$**)**.
Measured 12-Thread 5,760-Configuration Combined Impairment Sign-Off (reports/stage11_rf_matrix_coverage
Executing `.\scripts\run_stage11_matrix_parallel.bat 12 1 combined` in Xilinx Vivado 2022.2 `xsim` verified **all 5,760 / 5,760 valid CCSDS RF configurations (100.0% coverage, 0 failures)** under simultaneous `P5_COMBINED_MC` impairments in **343.81 s (5.73 min) wall-clock time**, transferring **11,520 RF packets** (`96,000 B` Uplink + `106,560 B` Downlink Space Packet payload across **6,551,040 16-bit RF I/Q samples**: `3,179,520` UL + `3,371,520` DL) and correcting **all 11,520 pre-FEC injected channel bit errors to 0 post-FEC bit errors (BER** $= 0.0$**, FER** $= 0.0$**)**.

6 Orbital Space Flight Channel Loopback Validation (Stage 12 Sign-Off)

In **Stage 12 (`tb/tb_stage12_space_channel_loopback.sv`)**, the IP core was validated in both **Spacecraft Flight (`ROLE_MODE=0`)** and **Ground EGSE (`ROLE_MODE=1`)** personas across our synthesizable **Orbital Space Flight Channel Model (`rtl/ccsds_space_flight_channel_model.sv`)** in **Xilinx Vivado 2022.2 `xsim`** (146,531,250 ps, 2,800 Uplink IQ samples, 4,336 Downlink IQ samples, **0 errors**):

Stage 12 Suite	Orbital Channel & Configuration Sweep	Measured Vivado 2022.2 `xsim` Result
`CFG-1`: 12 Waveforms × Variable Symbol Rates × 4-SPS Gardner	Sweeps all 12 PHY waveforms (BPSK, QPSK, OQPSK, PCM/PM × NRZ-L/M, Bi-ϕ-L), symbol rate dividers (div=1,2,4,8), and 4-SPS SRRC/Gardner timing offsets $\tau \in \{0, 1, 2, 3\}$.	**PASS:** Verified 28 multi-rate / multi-waveform / 4-SPS Gardner space channel configurations with **0 bit errors**.
`CFG-2`: Dynamic S-Curve Orbital Doppler & 4 Turnaround Ratios	Sweeps LEO horizon-to-horizon S-curve Doppler (+512 AOS → 0 TCA → −512 LOS, slew=−2/sample) + phase noise + FLL-Assisted Costas + S (240/221), X (880/749), Ka-B (3344/749) & Ka-A (3360/749) turnaround.	**PASS:** All 4 coherent turnaround ratios locked and tracked dynamic S-curve Doppler with **0 bit errors**.
`CFG-3`: Slant-Range Delay & Regenerative T4B/T2B PN Ranging	Tests two-way slant-range propagation delay (tau_prop = 4 & 9 samples) + Rician multipath + AWGN across Spacecraft (ROLE=0) and Ground EGSE (ROLE=1).	**PASS:** Both Spacecraft and Ground EGSE achieved T4B & T2B PN ranging lock (pn_All_Locked_out == 1).
`CFG-4`: All 5 CCSDS Protocols & Multi-FEC CADU Streams	Sweeps **TC**, **TM+OBT**, **AOS+FHEC**, **Proximity-1+CRC32**, and **USLP** across the space channel with BCH(63, 56) and Dual-Basis RS(255, 223) CADUs.	**PASS:** All 5 CCSDS protocols and Downlink CADU streams recovered with **0 errors**.
`CFG-5`: Mid-Pass Rician Deep Fade & COP-1 Go-Back-N Replay	Injects a mid-pass Rician deep fade burst erasing Frame N(S)=1 → FARM-1 asserts CLCW.Retransmit=1 (V(R)=1) → FOP-1 replays N(S)=1, 2.	**PASS:** Automatic Go-Back-N replay restored V(R)=3 with **0 lost or duplicate packets**.
`CFG-6`: Emergency HCPD (`VCID=63`) Over Space Channel	Transmits authenticated Emergency Command (MAGIC_OBC_HARD_RESET) over impaired space channel into u_spacecraft_top.	**PASS:** Decoded and asserted hcpd_obc_reset_pulse == 1 over space channel.

7 Future Enhancements Roadmap (Cataloged Only)

1. **Higher-Order Payload Downlink Modulations (`CCSDS 131.2-B-1 SCCC / 131.3-B-2 DVB-S2X`):** Optional GMSK, 8PSK, 16APSK, 32APSK, and 64APSK constellations for multi-Gbps X/Ka-band payload telemetry downlinks.
2. **Long-Block Downlink AR4JA LDPC ($k = 1024, 4096, 16384$) & Uplink** $(512, 256)$ **Table 3-1 ROM:** Expanding parity-check matrix ROMs for deep-space long-block LDPC and $(512, 256)$ uplink short-block LDPC.
3. **256-Bit Key Schedule (`AES-256-GCM`) & `HMAC-SHA-256`:** Extending ccsds_sdls_engine.sv from 10-round AES-128-GCM to 14-round AES-256-GCM and optional HMAC-SHA-256.
4. **Autonomous Scatter-Gather AXI4/AHB Master DMA & Foundry Rad-Hard Tech-Mapping:** Optional scatter-gather descriptor ring DMA master wrapper and vendor primitive techmap cells for RTG4, PolarFire RT, XQRKU060, and NanoXplore.